9.9.3. Seed Certification

Seed certification is a legally sanctioned system for quality control of seed multiplication and production. The purpose of this certification is to maintain the seeds and make them available to the public. Through certification, high quality seeds and propagating materials of notified kind and variety are grown and distributed to ensure genetic identity and purity.

Summary

Economic Botany deals with the relationship between people and economically important plants to fulfill the three basic needs of life such as food, clothing and shelter. Domestication, a term often used for a more intricate process, involves the genetic alteration of plants which did not appear at once, but rather over a substantial period of time, perhaps hundreds of years for some species. In the history of agriculture Vavilov has given the eight main centres of origin of plants were now divided into 12 centres of origin. In Organic agriculture biofertilizers are microbial inoculants which all ecofriendly, more effective even though cost effective than chemical fertilizers. *Rhizobium, Azolla,* VAM and sea weeds are used as fertilizers which increase the crop yield many fold.

Plant breeding is a purposeful manipulation of plant species in order to create desirable genotype and phenotype for the benefit of mankind. Plant introduction, selection, hybridization, heterosis, mutation breeding, polyploidy breeding and green revolution are the different methods of conventional breeding.

Seed is an important part of the plant as it gives the food for future generation, so it should be carefully protected and stored. Seed hardening, seed treatment, seed pelleting, seed coating and bio priming of seeds are the modern methods of seed protection. Seeds are stored in bamboo structure, mud or earthen structure in conventional method of storage. In villages, the farmers store the entire seeds in concrete / cement silos or metal or plastic drums. In modern methods of storage we have cryo preservation, gene bank and Svalbard seed bank for long time seed storage.

Evaluation

UBP8Q

1. **Assertion:** Genetic variation provides the raw material for selection

   Reason: Genetic variations are differences in genotypes of the individuals.

   a) Assertion is right and reason is wrong.
   b) Assertion is wrong and reason is right.
   c) Both reason and assertion is right.
   d) Both reason and assertion is wrong.

2. While studying the history of domestication of various cultivated plants _____ were recognized earlier

   a) Centres of origin
   b) Centres of domestication
   c) Centres of hybrid
   d) Centres of variation

3. Pick out the odd pair.

   a) Mass selection - Morphological characters
   b) Purline selection - Repeated self pollination
   c) Clonal selection - Sexually propagated
   d) Natural selection - Involves nature

4. Match Column I with Column II

Column I	Column II
i) William S. Gaud	I) Heterosis
ii) Shull	II) Mutation breeding
iii) Cotton Mather	III) Green revolution
iv) Muller and Stadler	IV) Natural hybridization

   a) i – I, ii – II, iii – III, iv – IV
   b) i – III, ii – I, iii – IV, iv – II
   c) i – IV, ii – II, iii – I, iv – IV
   d) i – II, ii – IV, iii – III, iv – I

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